BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

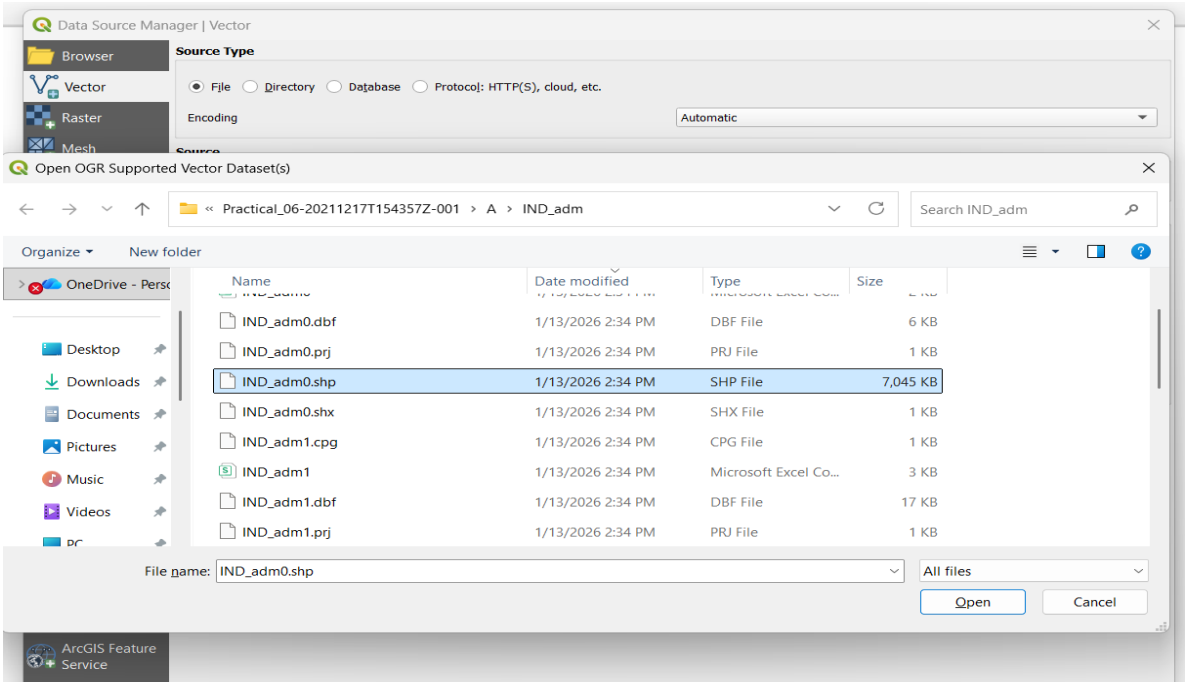
Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 28-01-2026**

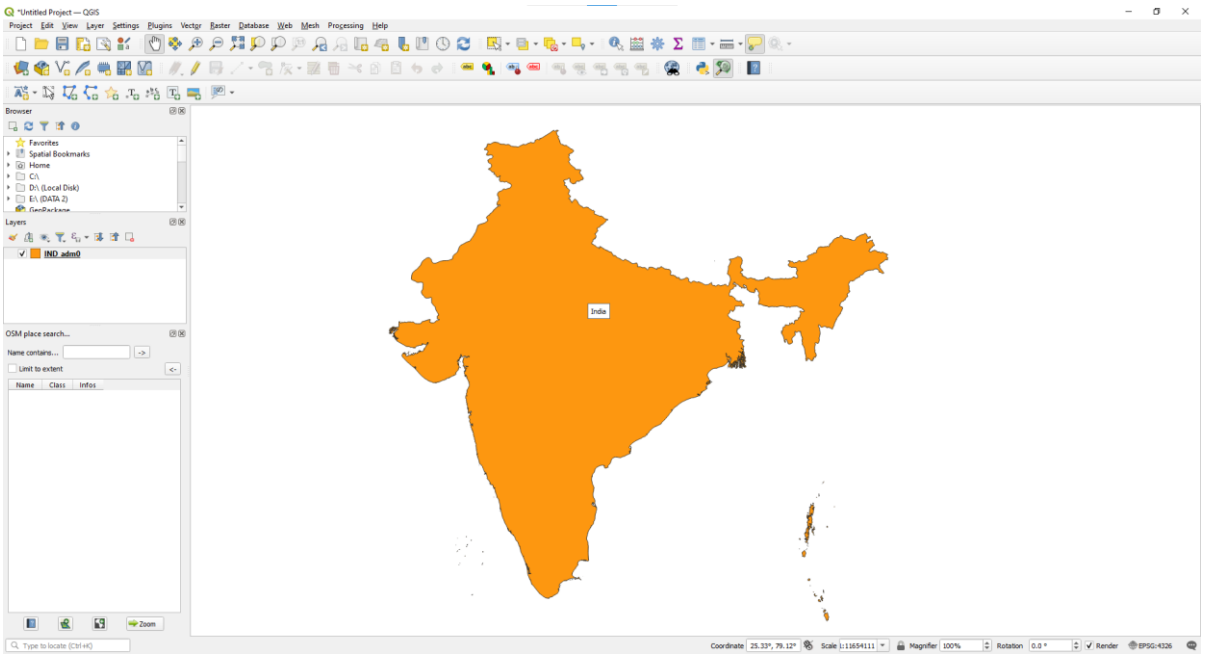
Course Name: Geographic Information System (GIS) **Page no:**

Practical Number:- 06

A) Georeferencing Topo Sheets and Scanned Maps.

- Start a new project
- Go to Layers → Add Layer → Add vector Layer
- Select GIS_Workshop\Manual\Prac06\IND_adm0.shp





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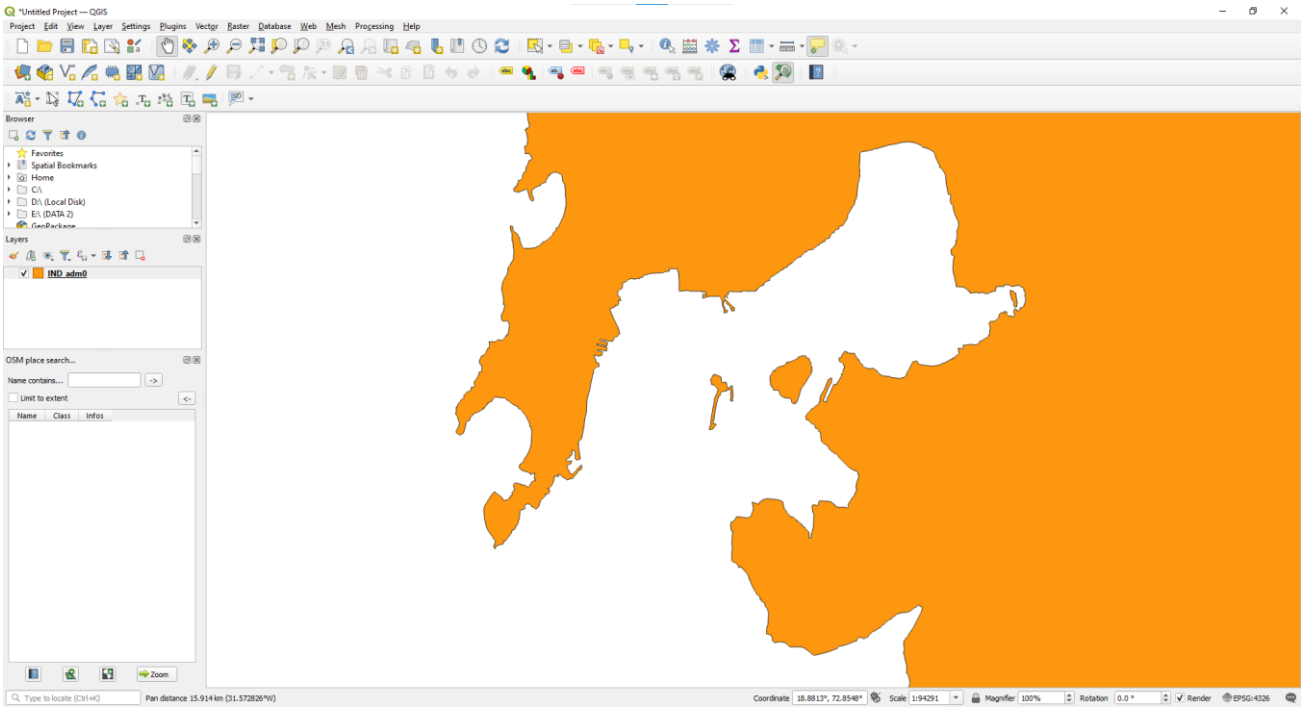
BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 28-01-2026**

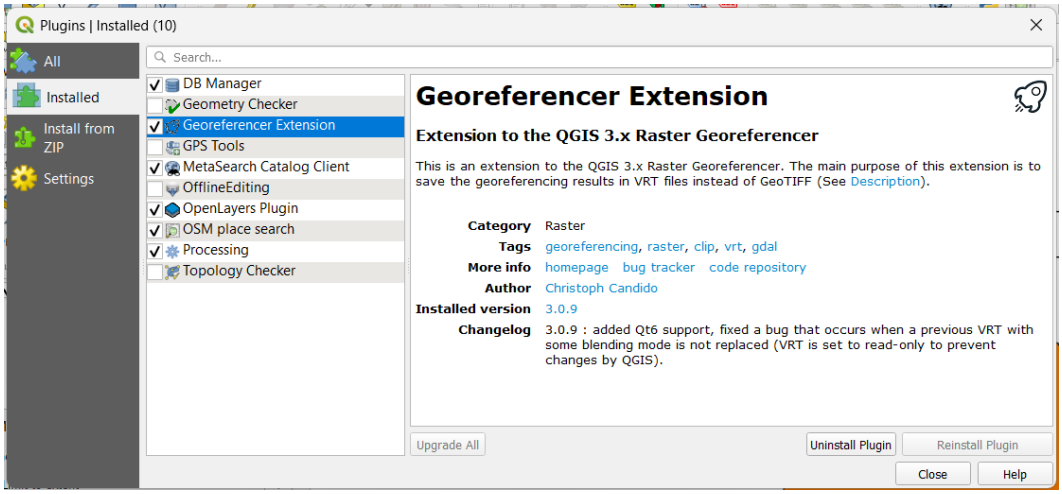
Course Name: Geographic Information System (GIS) **Page no:**

Practical Number:- 06

- Zoom in to Mumbai region in the layer.



- Go to Plugins→ Manage and Install Plugins
- Install Georeferencer Extension



- Ensure that is GeoReferencer GDAL is checked, if not install Georeferencer GDAL plugin.
- Go to Raster → Georeferencer

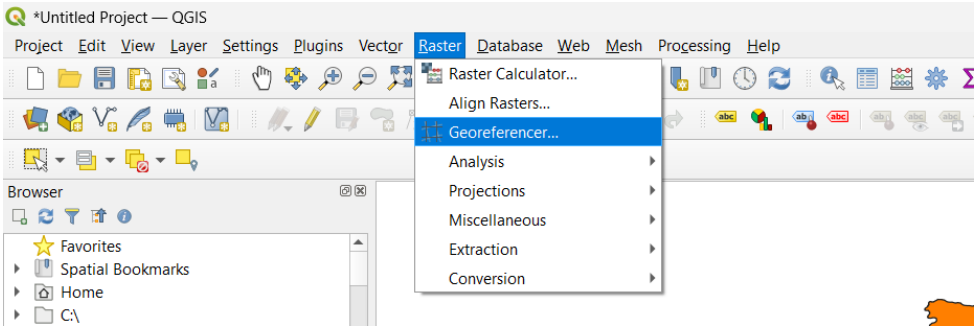
Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

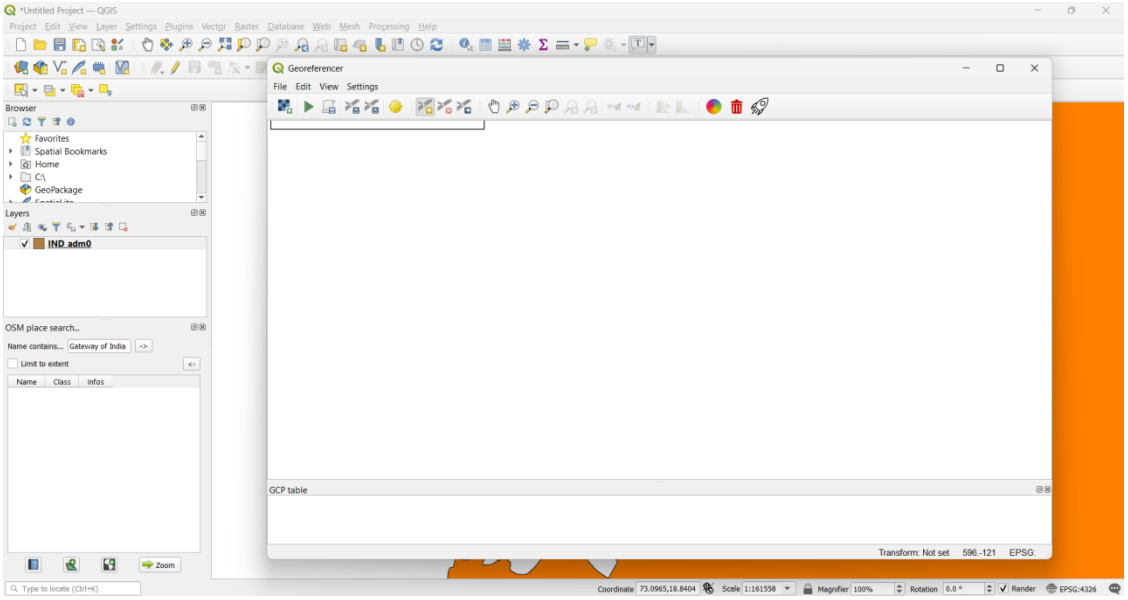
Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 28-01-2026**

Course Name: Geographic Information System (GIS) **Page no:**

Practical Number:- 06



- A new Georeferencer window will open



- File → Open Raster

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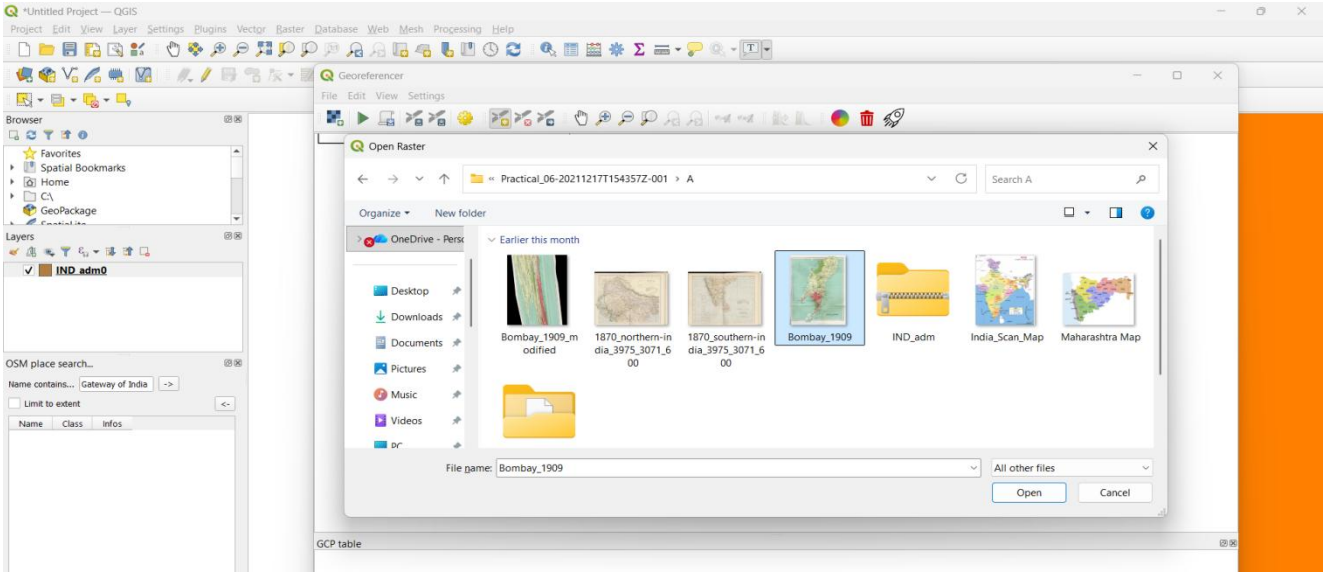
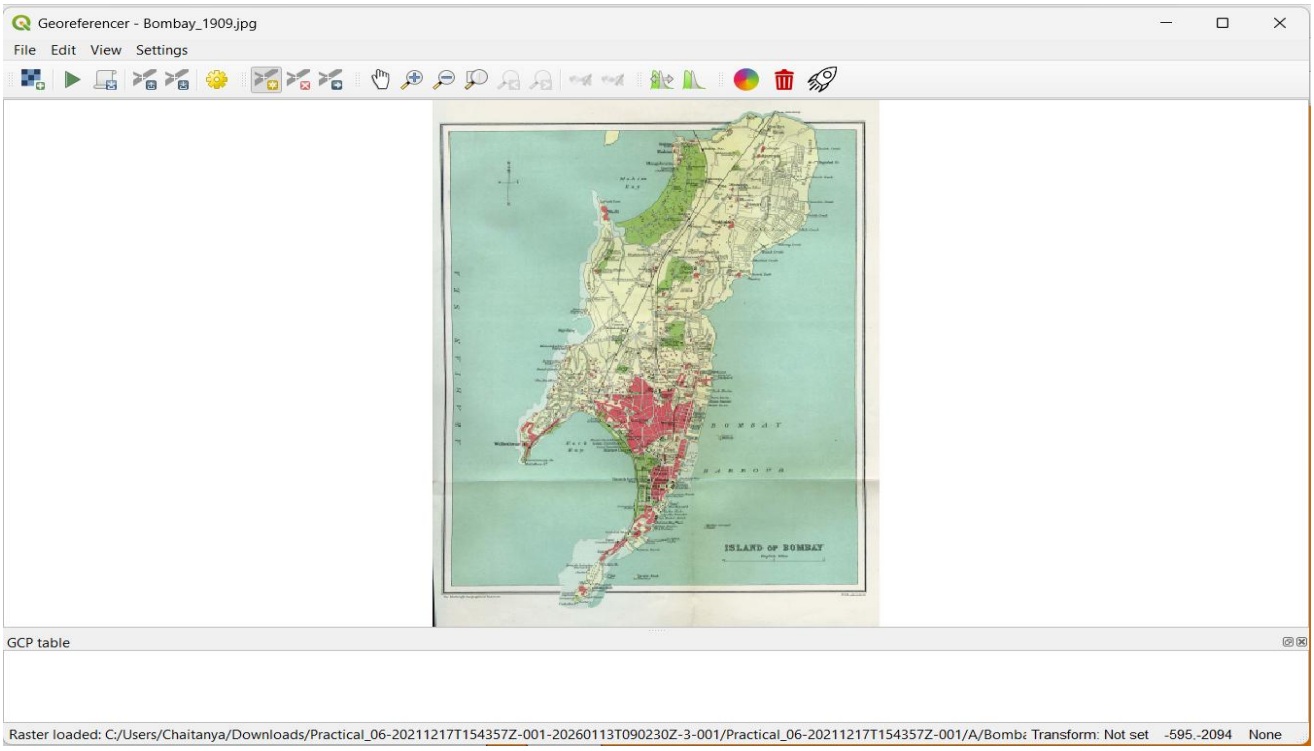
BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 28-01-2026**

Course Name: Geographic Information System (GIS) **Page no:**

Practical Number:- 06





- Go to Settings →Transformation Settings

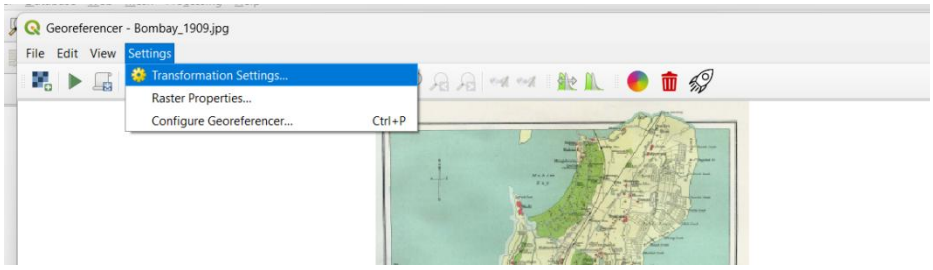
Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

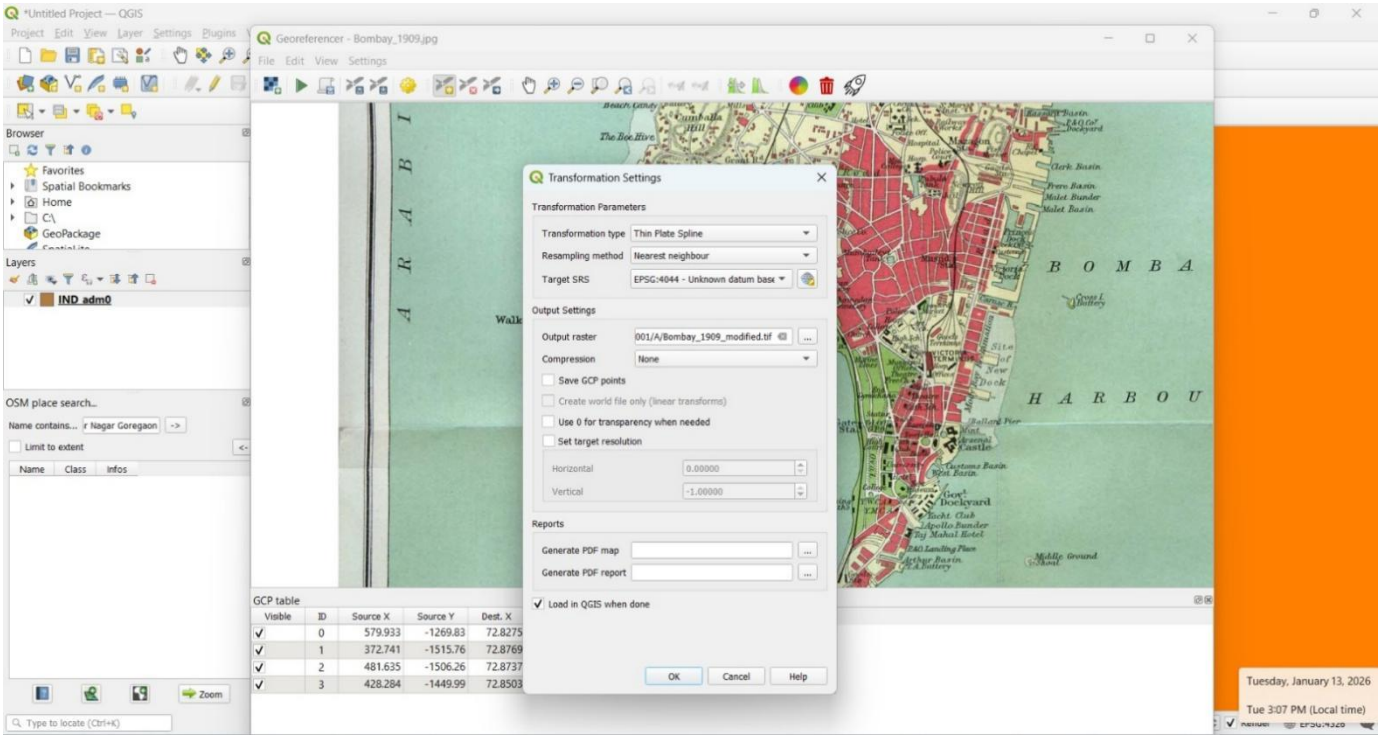
Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 28-01-2026**

Course Name: Geographic Information System (GIS) **Page no:**

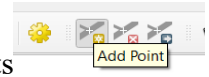
Practical Number:- 06



- In the Transformation Settings window



- Select Transformation type → Thin Plate Spline
- Re-sampling Method → Nearest Neighbour ▪
- Target TRS → Everest 1830 datum: EPSG 4044
- Select Output Raster Name and Location
- Check the Load in QGIS When Done Option
- Press "OK".

- In Georeferencer window Go to Edit → Add Points 

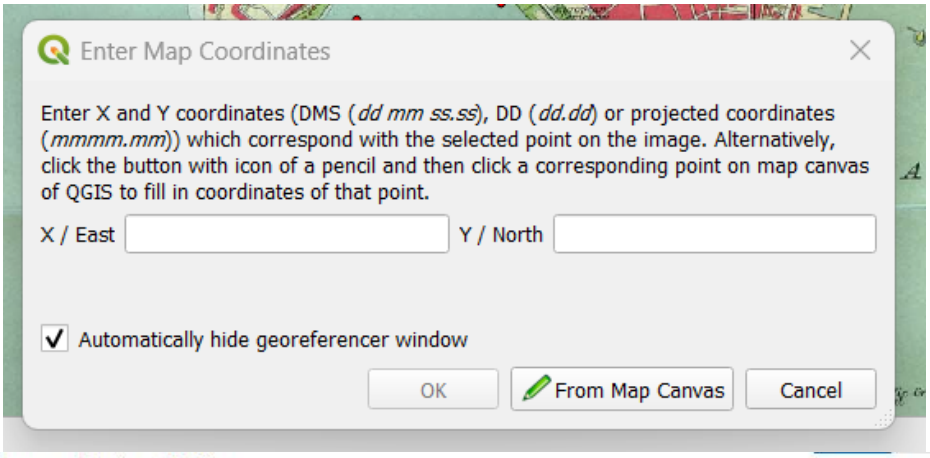
Teacher's Signature

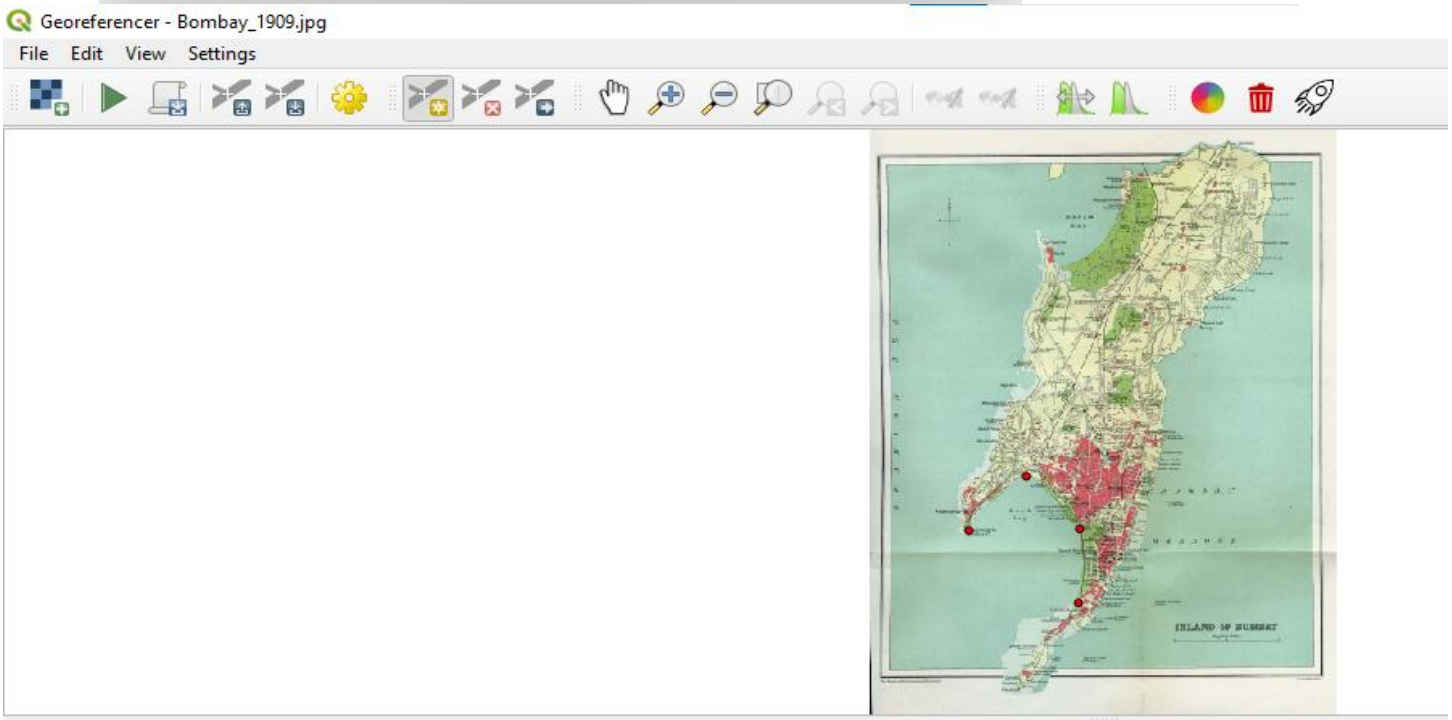
BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 28-01-2026**

Course Name: Geographic Information System (GIS) **Page no:**

Practical Number:- 06





GCP table

Visible	ID	Source X	Source Y	Dest. X	Dest. Y	dX (pixels)	dY (pixels)	Residual (pixels)
✓	0	329.502	-1439.96	72.794	18.939	1.7053e-13	2.27374e-13	2.84217e-13
✓	1	523.154	-1245	72.8104	18.9546	2.27374e-13	2.27374e-13	3.21555e-13
✓	2	703.721	-1434.07	72.8235	18.9338	2.27374e-13	2.27374e-13	3.21555e-13
✓	3	699.142	-1699.69	72.8157	18.9146	2.27374e-13	0	2.27374e-13

- Select the set of control points.
- Go to, Setting → transformation settings.

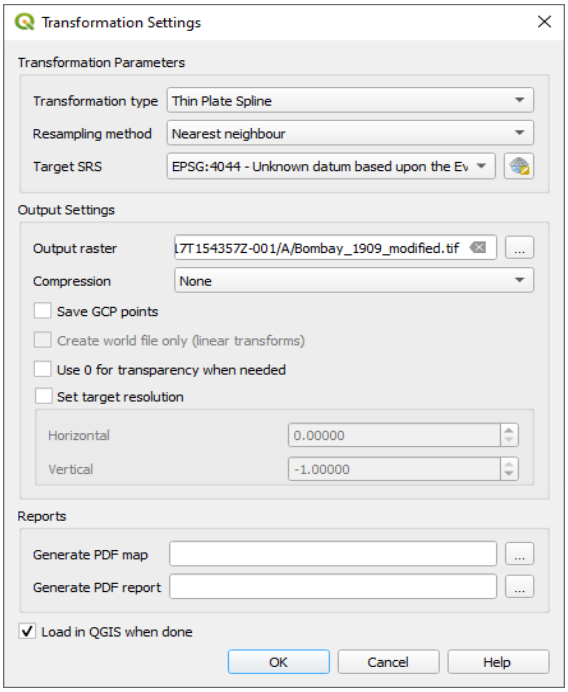
Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

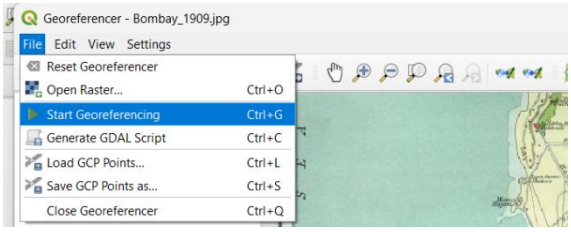
Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 28-01-2026**

Course Name: Geographic Information System (GIS) **Page no:**

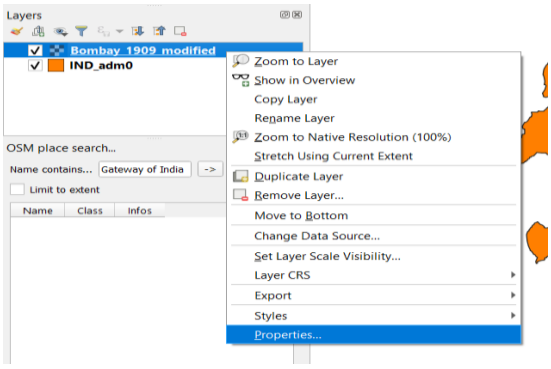
Practical Number:- 06



- Press "RUN"
- In Georeferencing window go to → File → Start Georeferencing



- The progress indicator will appear
- The canvas area will now have the scanned map of Mumbai referenced with control points.
- Select the newly added layer in Layer Panel Right click and go to property.



Teacher's Signature

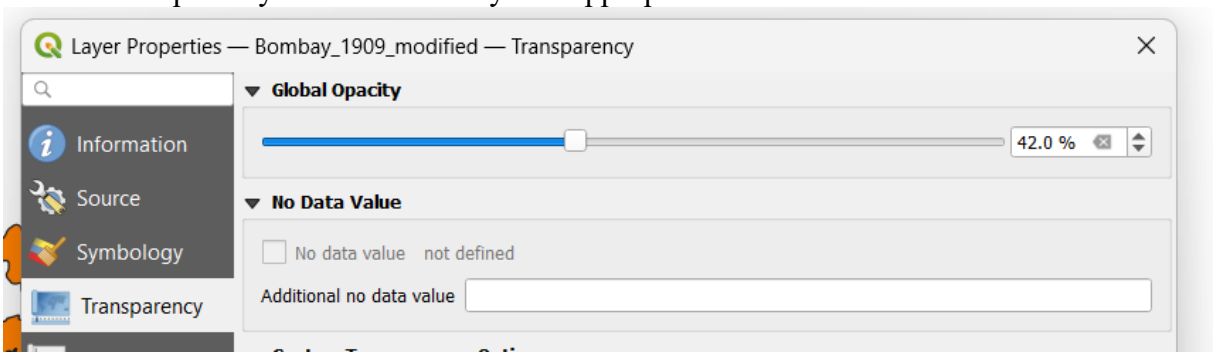
BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 28-01-2026**

Course Name: Geographic Information System (GIS) **Page no:**

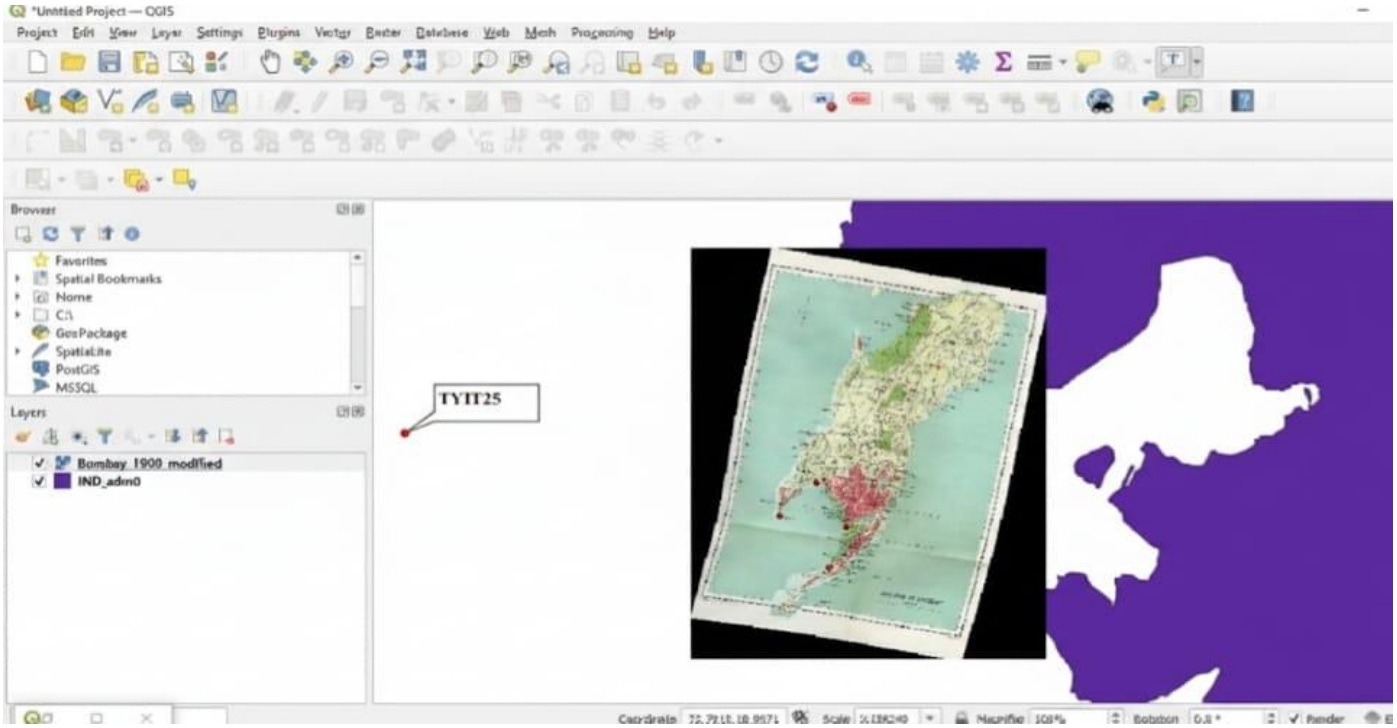
Practical Number:- 06

- Set Transparency level of raster layer to appropriate level.



Output:

The Scanned Image map coincides with the existing map.



Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

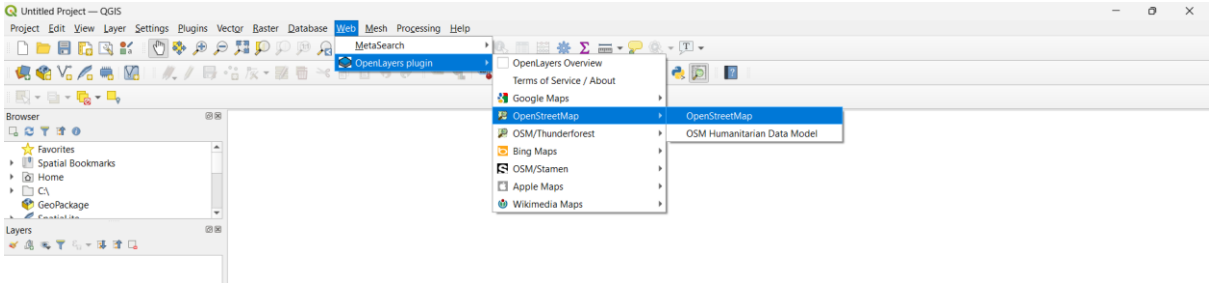
Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 28-01-2026**

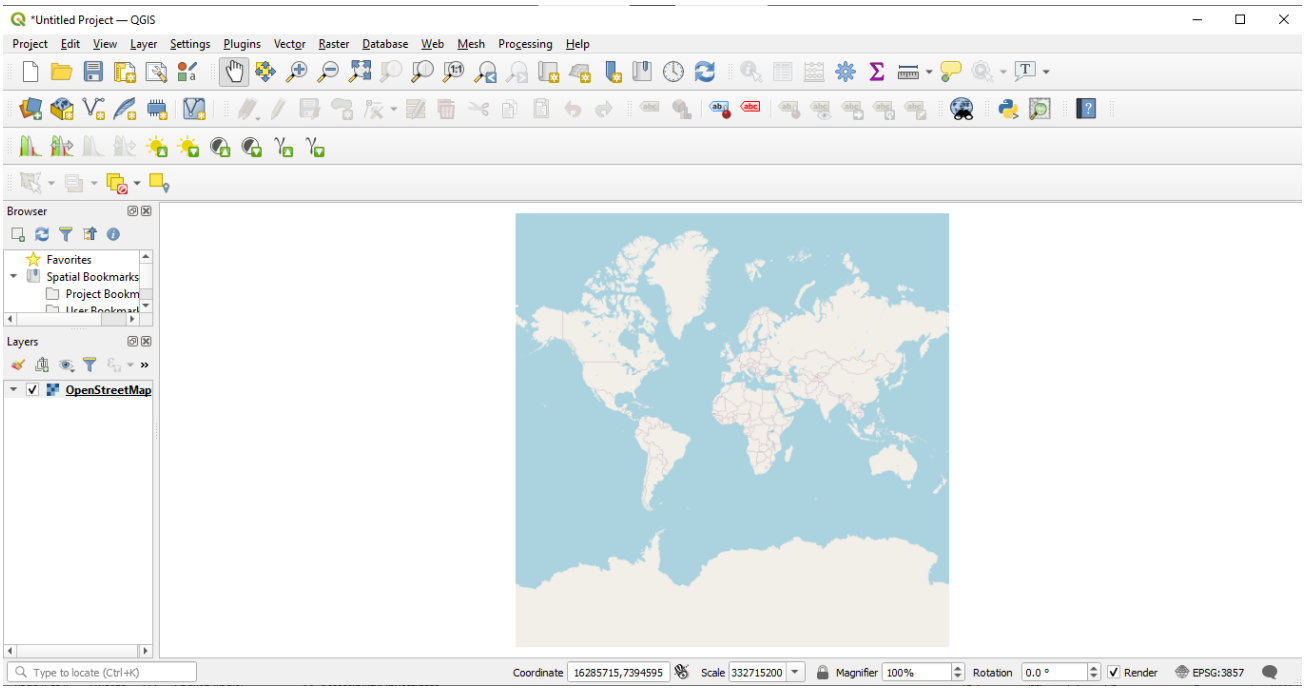
Course Name: Geographic Information System (GIS) **Page no:**

Practical Number:- 06

B) Georeferencing Aerial Imagery

- Install plugin OpenStreetMap
- Go to Web Menu → OpenLayerPlugin → OpenStreetMap→ OpenStreetMap





- Go to Project → Properties → Set CRS to EPSG 3857

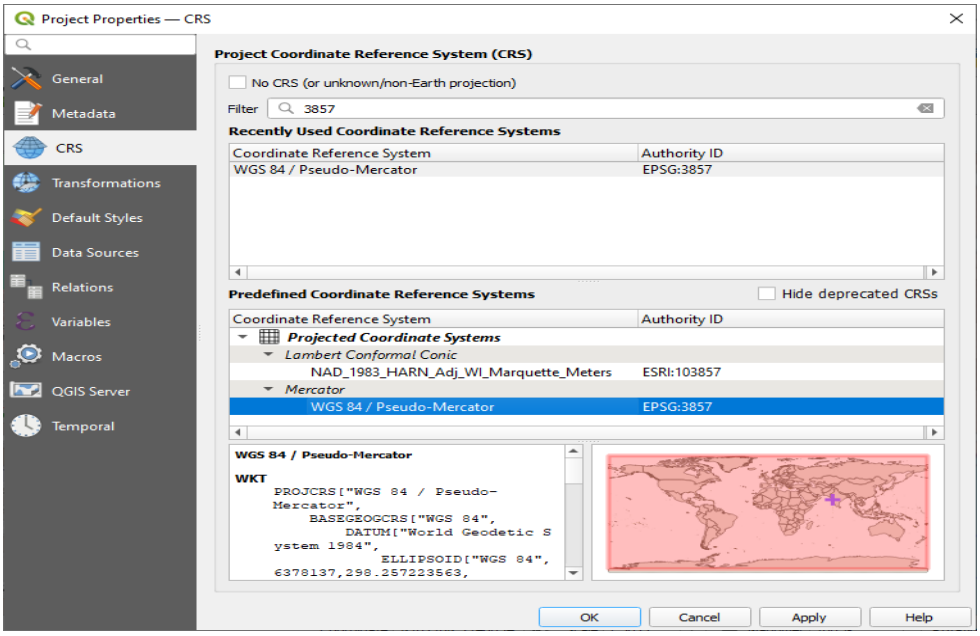
Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

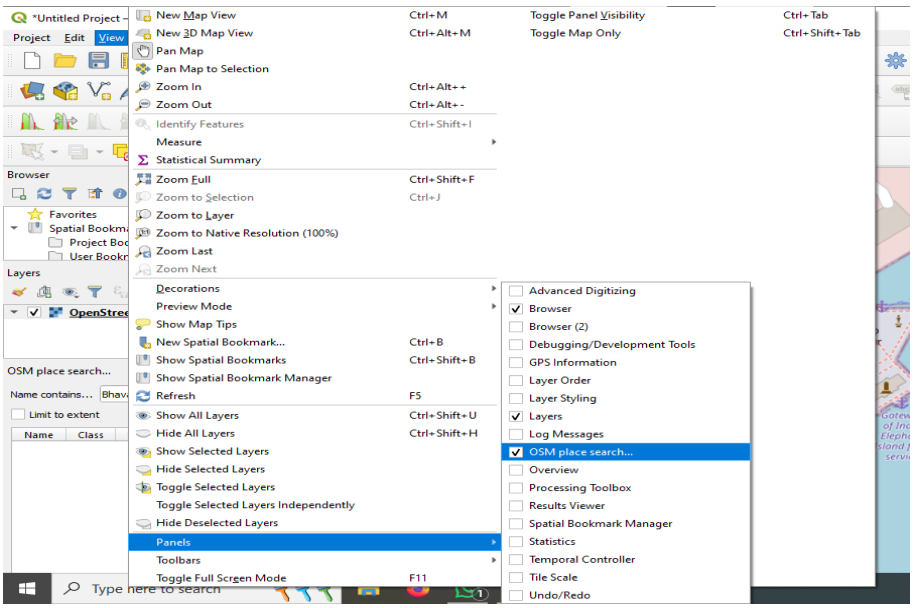
Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 28-01-2026**

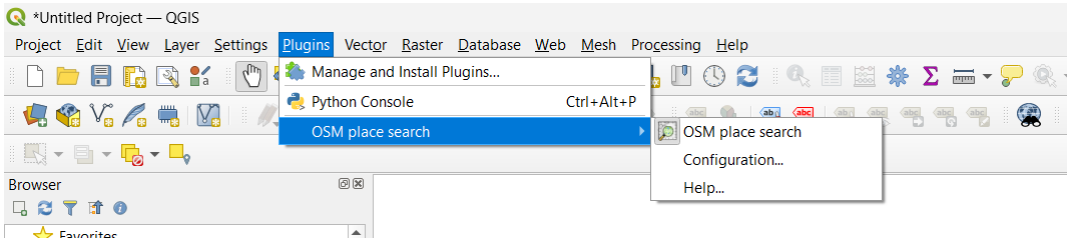
Course Name: Geographic Information System (GIS) **Page no:**

Practical Number:- 06



- Go to View → Panels → select OSM Place search
- Or From The Plugins → select OSM Place search





Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

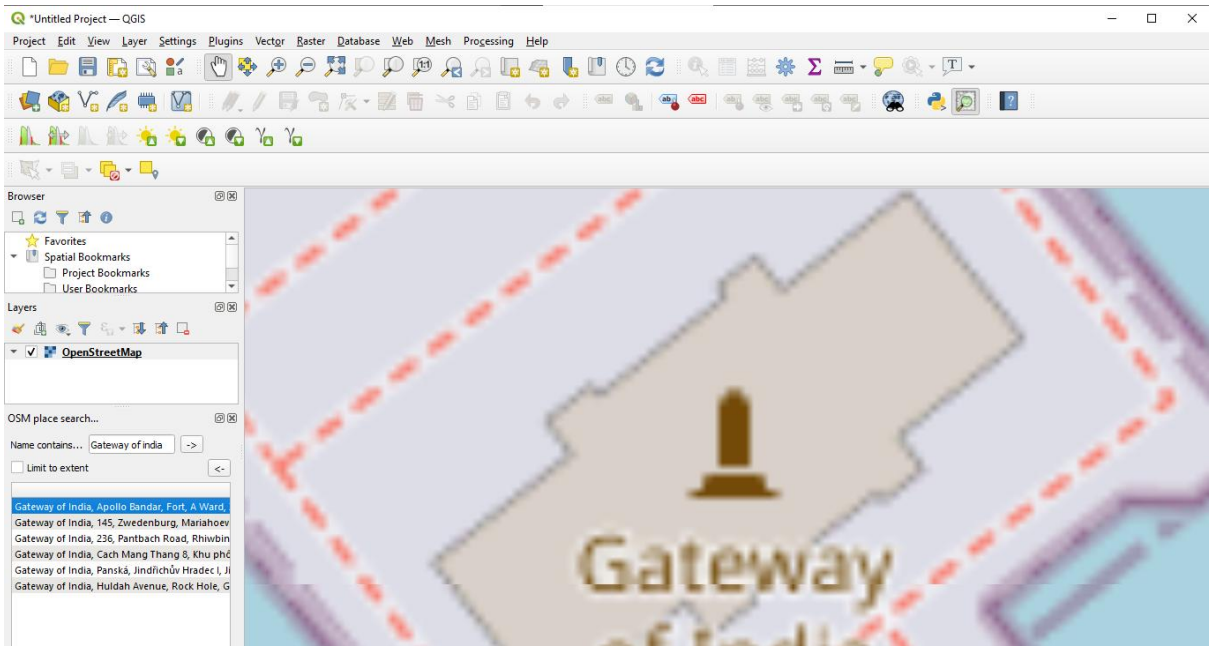
Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 28-01-2026**

Course Name: Geographic Information System (GIS) **Page no:**

Practical Number:- 06

- Select The Location The Gateway of India, Mumbai is located at 18.92°N 72.83°E
- Search Gateway of India in OSM Search Panel

- Zoom in to appropriate level.
- The map will appear like this



- Go to Raster → Georefrencer
- A new Georeferencer window will open

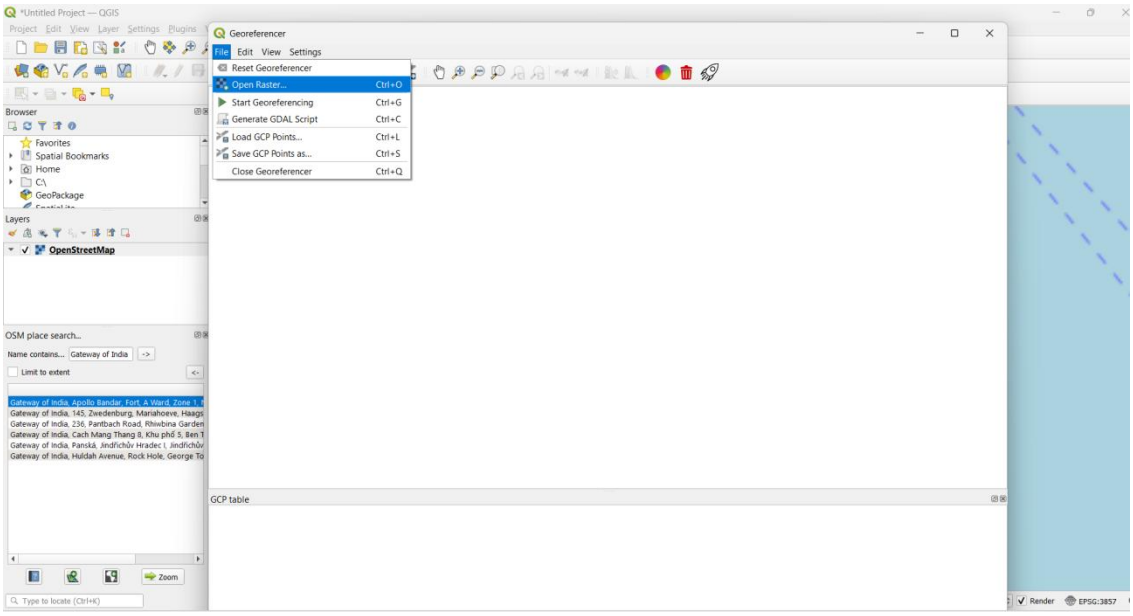
Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

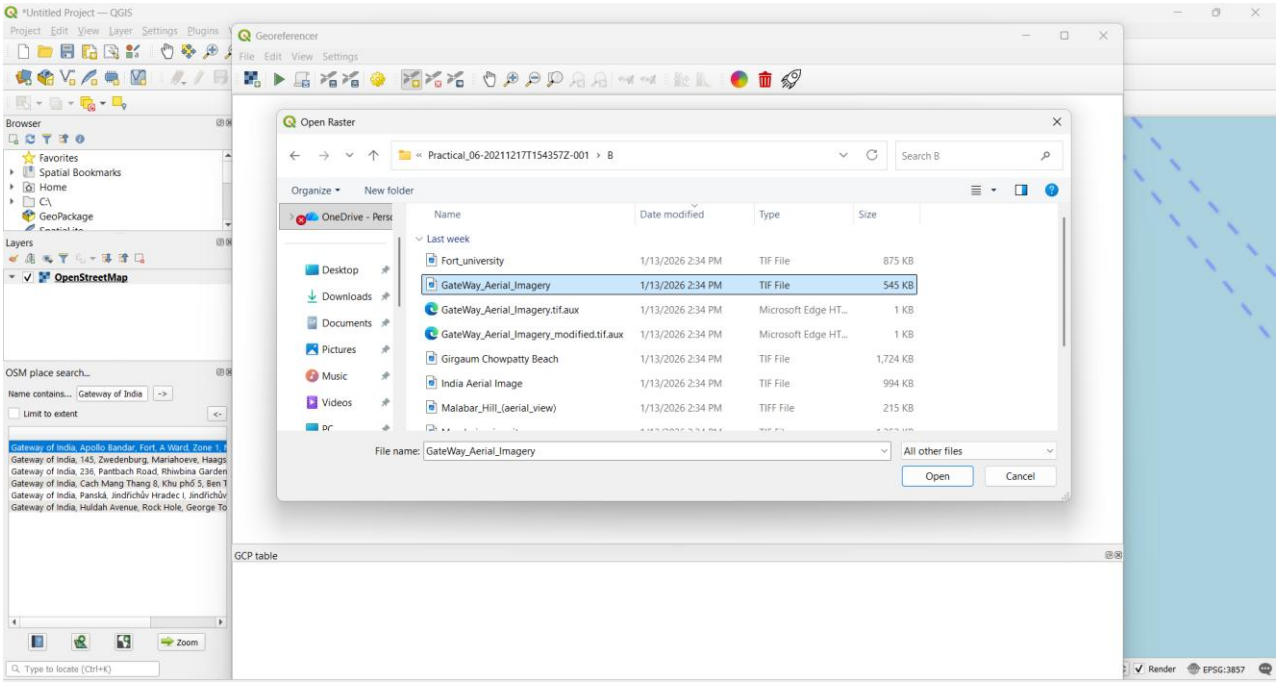
Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 28-01-2026**

Course Name: Geographic Information System (GIS) **Page no:**

Practical Number:- 06



- File → Open Raster
- Select file "Gateway_Imagery.tif" from project data folder



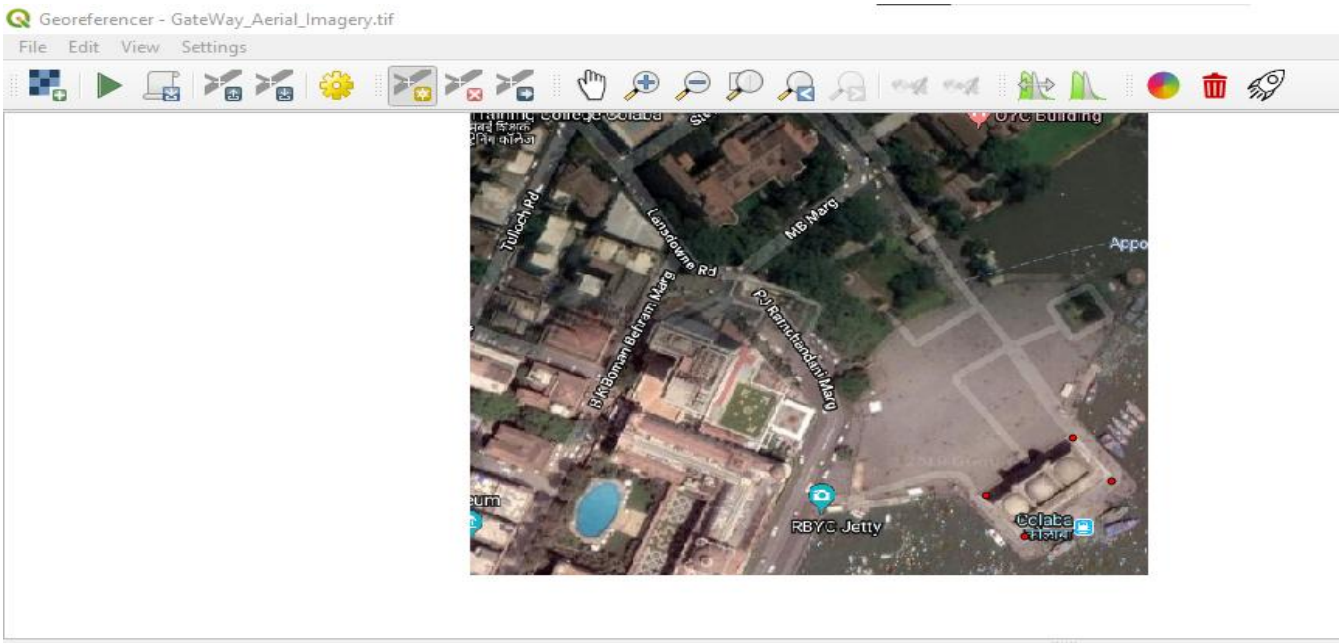
Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 28-01-2026**

Course Name: Geographic Information System (GIS) **Page no:**

Practical Number:- 06

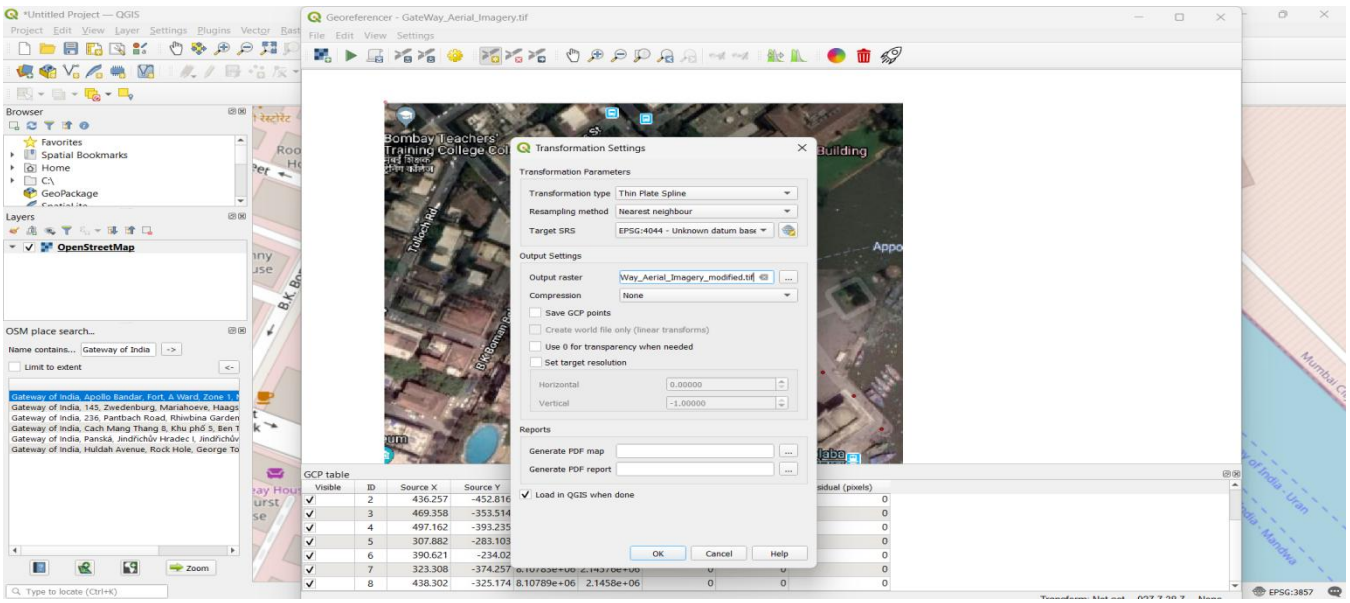


GCP table

Visible	ID	Source X	Source Y	Dest. X	Dest. Y	dX (pixels)	dY (pixels)	Residual (pixels)
✓	0	393.42	-401.715	8.10787e+06	2.14575e+06	0	0	0
✓	1	459.78	-349.575	8.10792e+06	2.14578e+06	0	0	0
✓	2	423.045	-438.45	8.10789e+06	2.14572e+06	0	0	0
✓	3	489.405	-388.68	8.10794e+06	2.14576e+06	0	0	0

- Go to Edit → Add Point
- Select control points from map

Go to Setting → Transformation Setting



Teacher's Signature

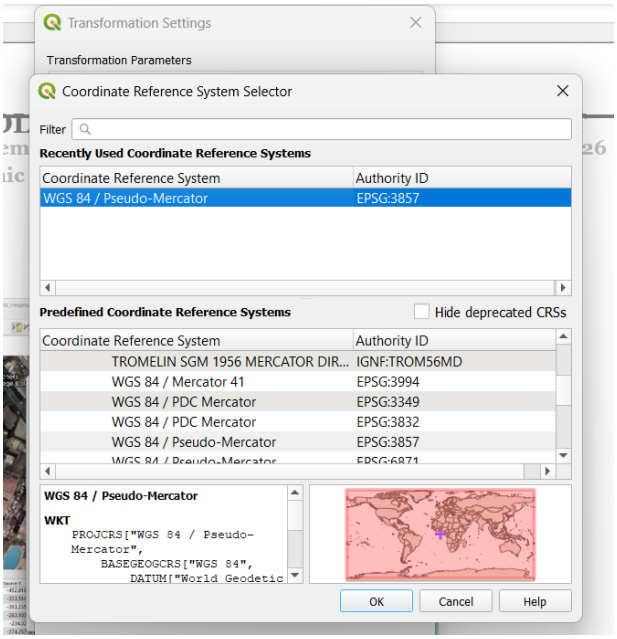
BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 28-01-2026**

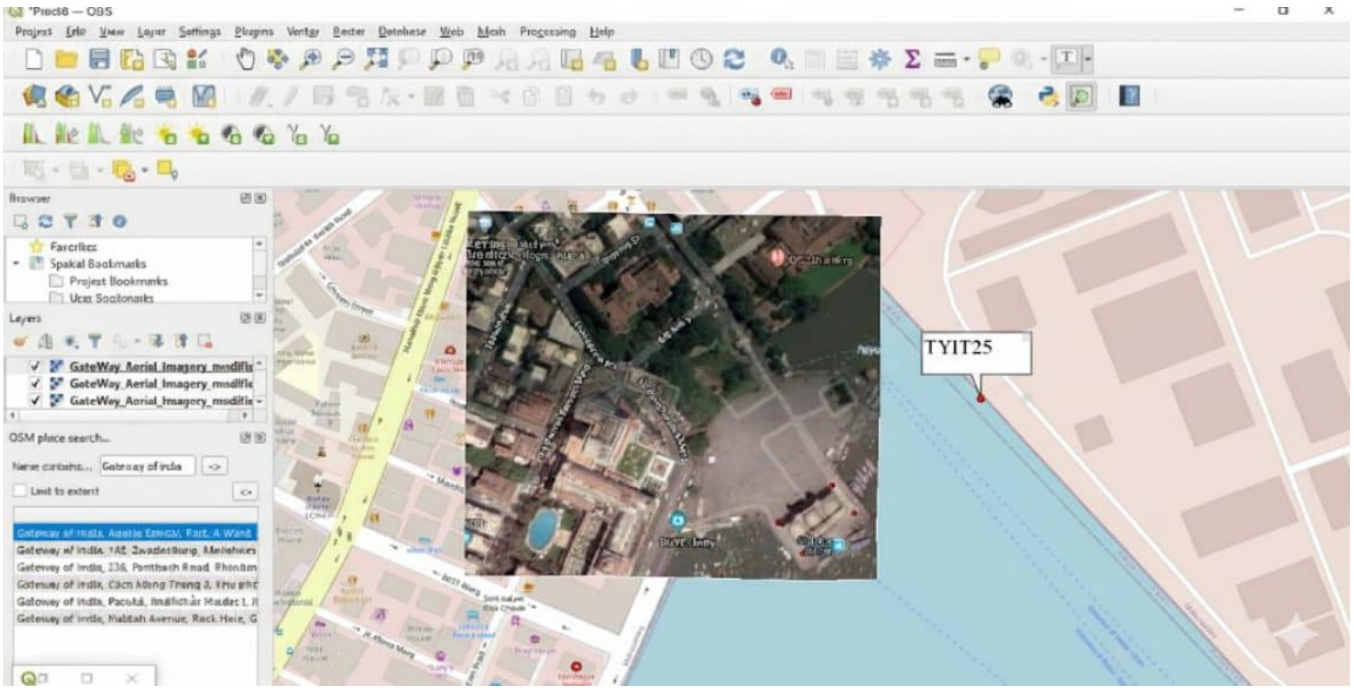
Course Name: Geographic Information System (GIS) **Page no:**

Practical Number:- 06

- Select The EPSG:3857 in Target SRS



- Go to File → Start Georeferencing or Press the button in Georegerencing Window.
- Observe that the aerial image of the Gateway of India is georeferenced on OSM in the map canvas.
- Here is The Final Result:



Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

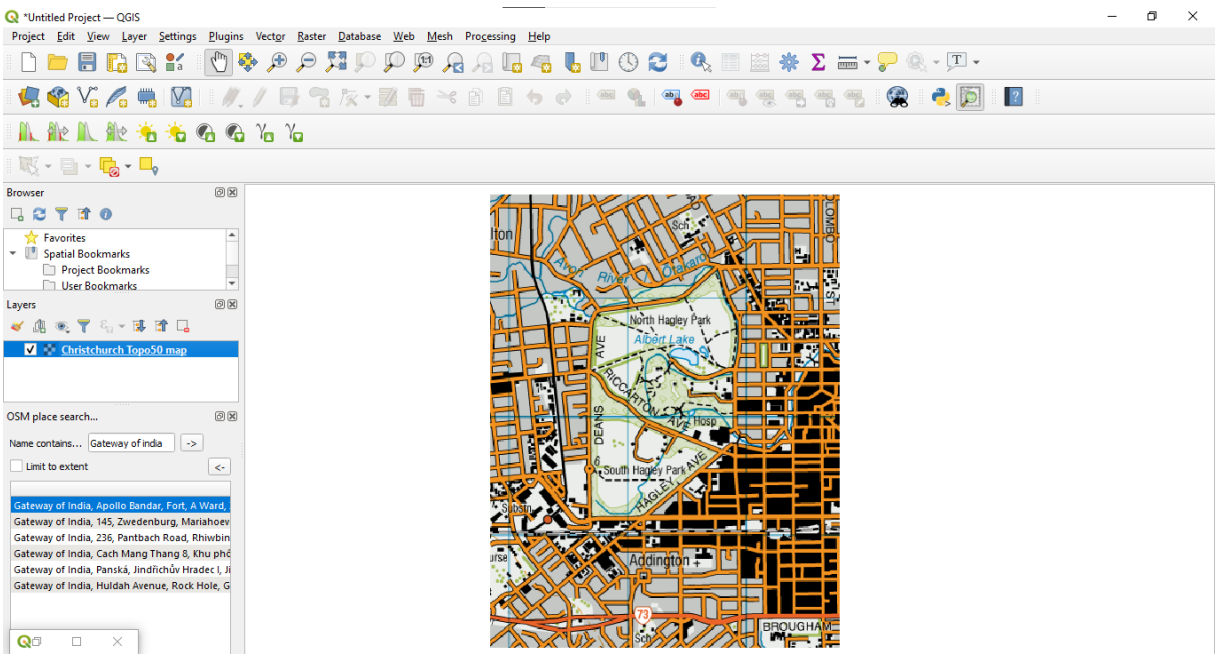
Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 28-01-2026**

Course Name: Geographic Information System (GIS) **Page no:**

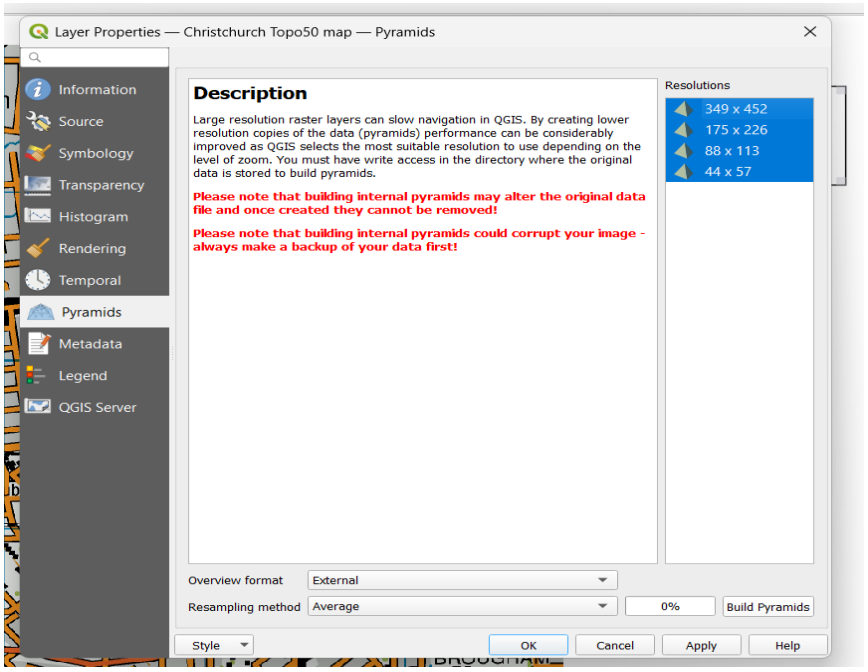
Practical Number:- 06

C) Digitizing Map Data

- Go to Layer ‣ Add Raster→ Select "Christchurch Topo50 map.tif" from project Folder.



- Right-click the Christchurch Topo50 map.tif layer and select Properties.
- Choose the Pyramids tab. Hold the Ctrl key and select all the resolutions offered in the Resolutions panel.



Teacher's Signature

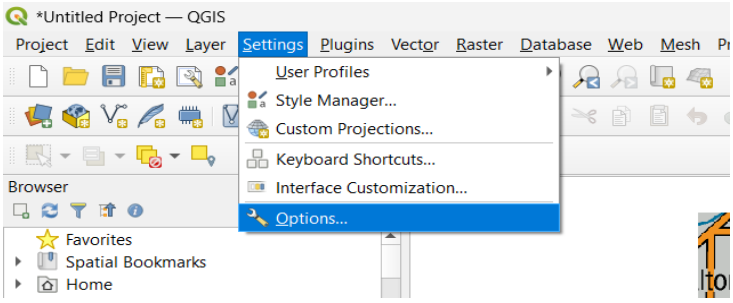
BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 28-01-2026**

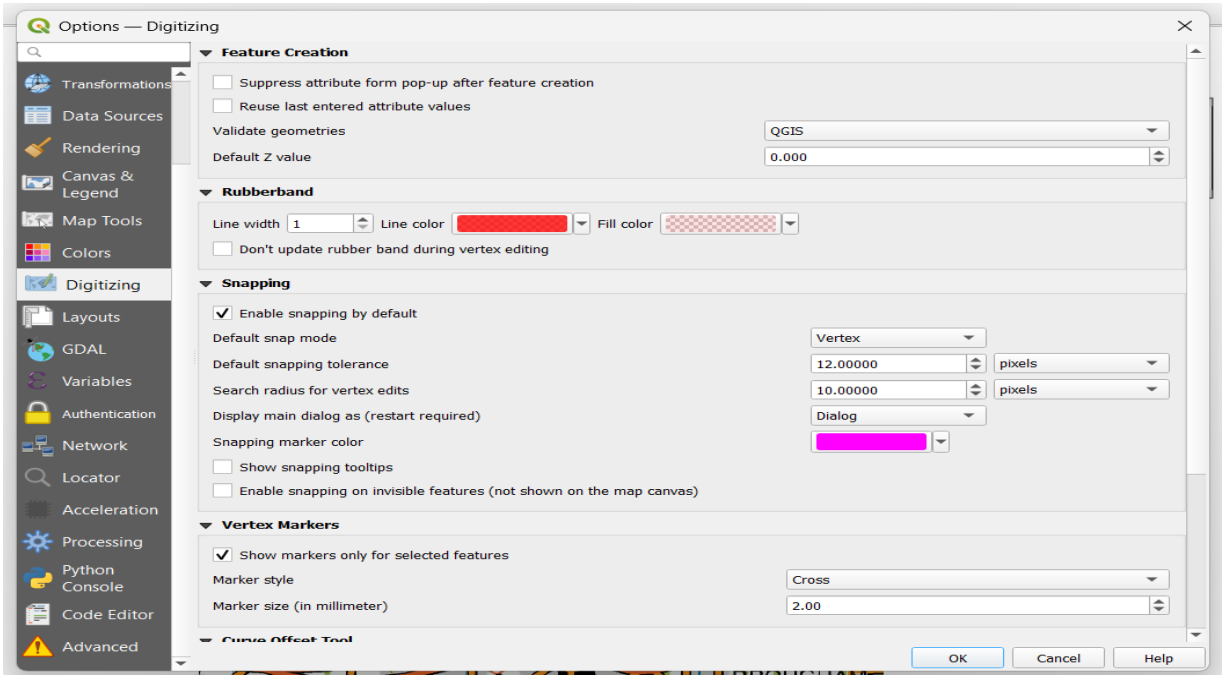
Course Name: Geographic Information System (GIS) **Page no:**

Practical Number:- 06

- Click Build pyramids. Then click OK.
- Go to Settings →Options.... Select the Digitizing tab in the Options dialog.



- Set the Default snap mode to vertex and segment.
- Do Changes According to it in below Categories



- Press OK.
- Go to Layer → Add Layer → Add Spatialite Layer.
- Select the name and location for Spatial database eg: "GIS_Workshop\Practicals\Practical_06\C\Digitizeroad.sqlite".
- Name the Layer as "Digitized_Road
- Set Geometry type as "Line"
- Set CRS EPSG:4167 – NZGD2000

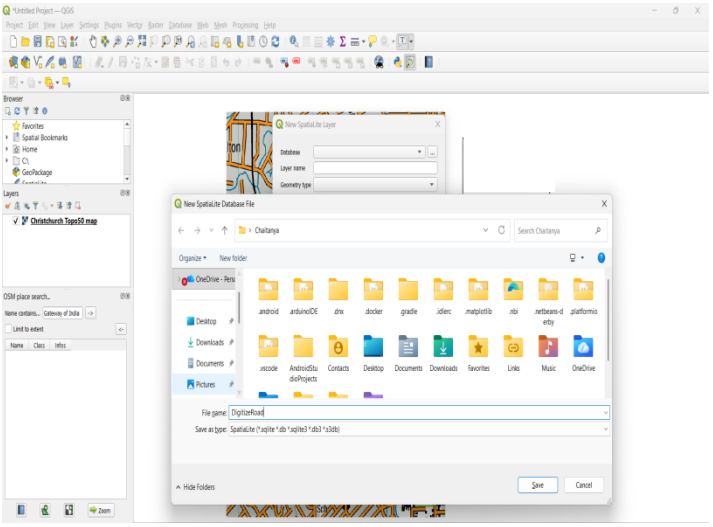
Teacher's Signature

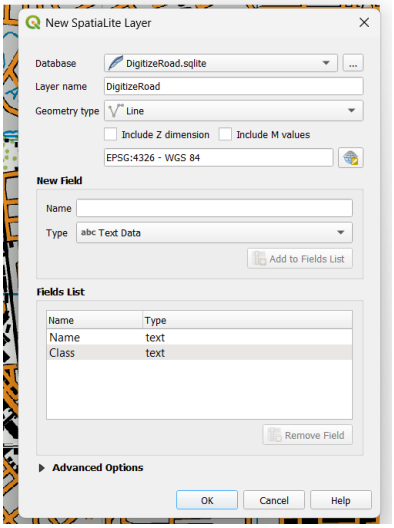
BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 28-01-2026**

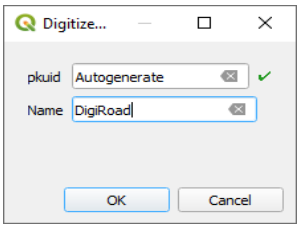
Course Name: Geographic Information System (GIS) **Page no:**

Practical Number:- 06

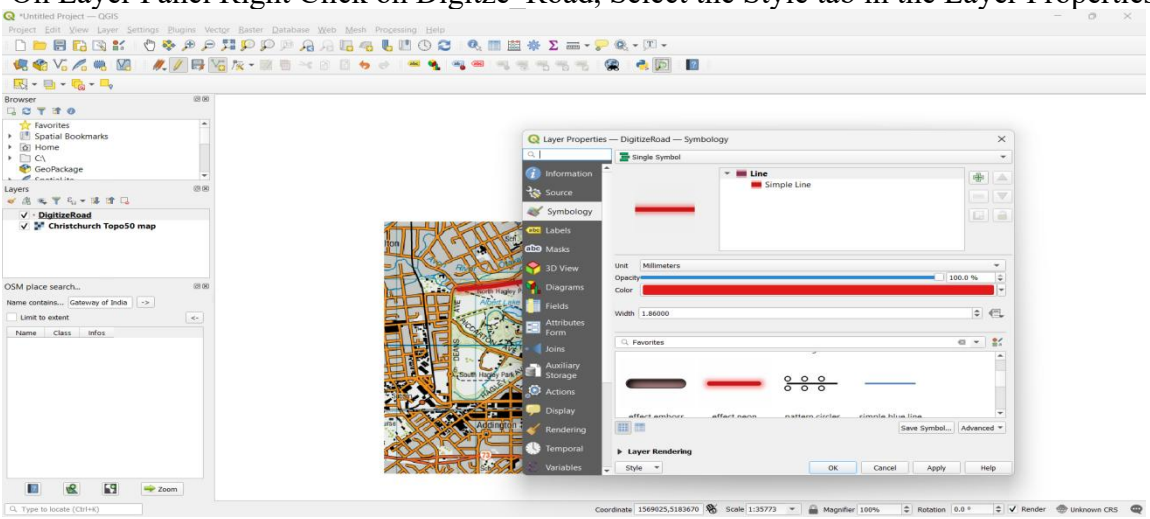




- Add "Name" and "Class" fields using "Add to Fields List".
- Once the layer is loaded, click the Toggle Editing button to put the layer in editing mode.
- Click the Add feature button. Click on the map canvas to add a new vertex. Add new vertices along the road feature. Once you have digitized a road segment, right-click to end the feature.



- On Layer Panel Right Click on Digitze_Road, Select the Style tab in the Layer Properties dialog.



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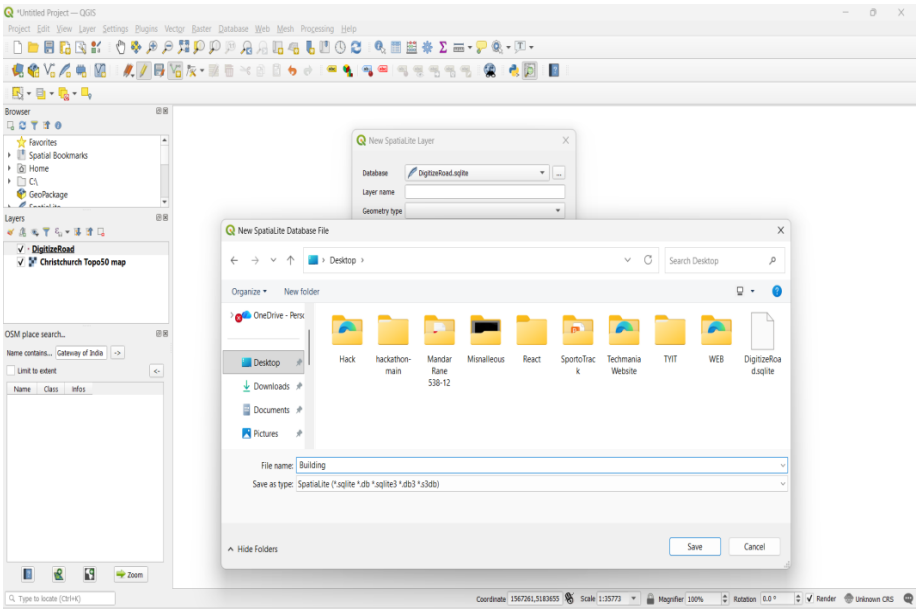
PRACTICAL JOURNAL

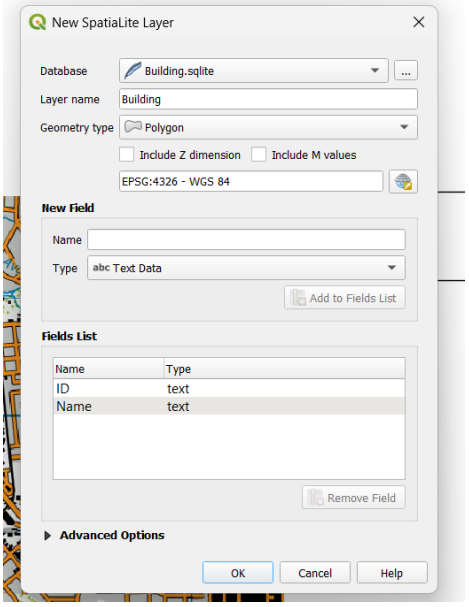
Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 28-01-2026**

Course Name: Geographic Information System (GIS) **Page no:**

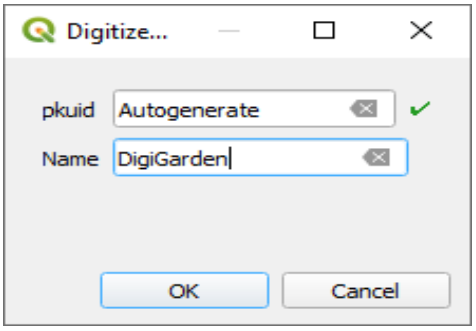
Practical Number:- 06

- Select appropriate style to see the digitized road feature clearly.
- Same as it is create a layer for building





- Select Digitized_Building layer in Layer Panel and click on Toggle Editing button and then Add Polygon Feature button on Tool bar.
- Add One Building to the region by adding polygon.



- The Layer will appear on map canvas
- Go to properties of Building Layer and Add The Single Label to Display The Name on the Layer

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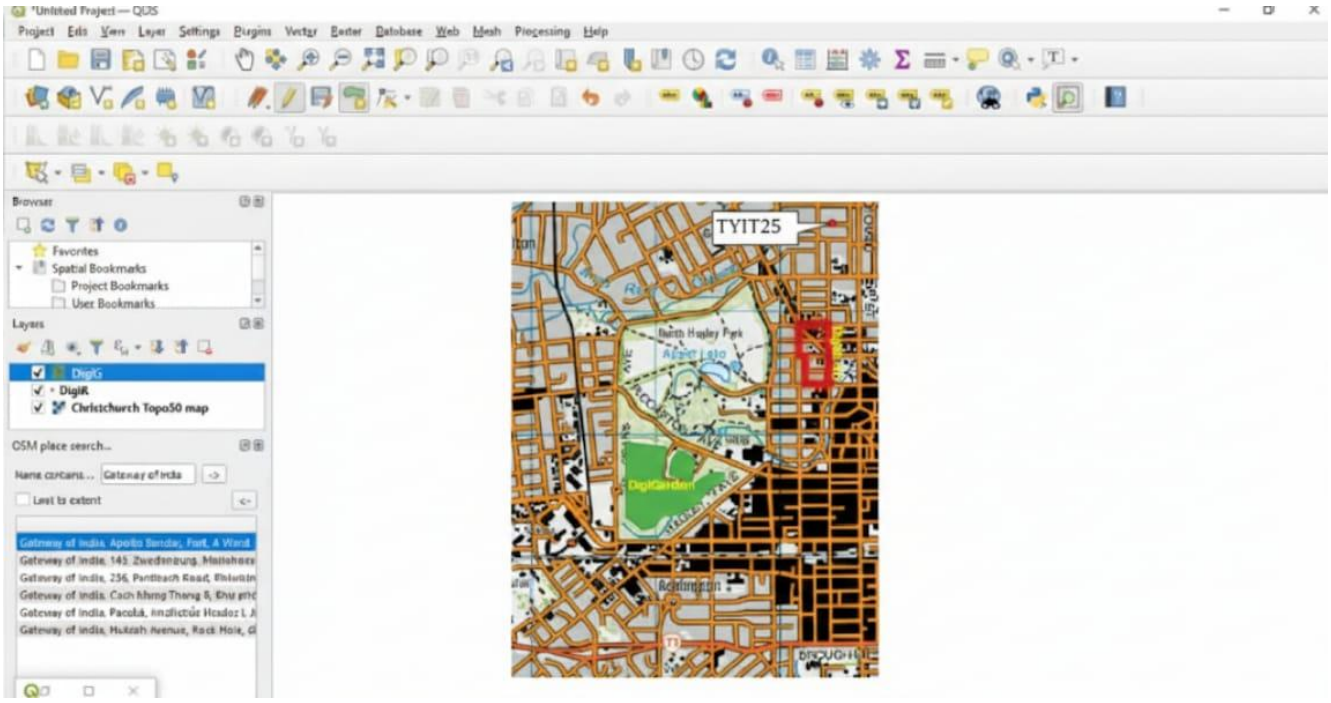
Course Name: Geographic Information System (GIS) **Page no:**

Practical Number:- 06



- The digitizing task is now complete. You can play with the styling and labeling options in layer properties to create a nice looking map from the data you created.

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